Business Problem for Supply Chain Inventory Dataset

A retail company manages multiple warehouses and suppliers. The company wants to **monitor inventory levels, supplier efficiency, and product demand** to ensure smooth supply chain operations.

The company needs insights to answer:

- Which **warehouse holds the most stock**
- Which **supplier delivers fastest**
- Which **products have the highest demand**
- Whether **stock levels are sufficient compared to orders**

The objective is to **optimize inventory distribution, reduce delivery delays, and ensure product availability**.

Key Performance Indicators (KPIs):

1. Inventory KPIs

Total Stock Quantity
Total available inventory across warehouses.

Total Order Quantity
Total demand from customers.

Stock vs Order Ratio
Shows if inventory is enough to meet demand.

Average Stock per Product

2. Supplier KPIs

Average Delivery Time

Fastest Supplier

Orders by Supplier
Total order quantity supplied by each supplier.

3. Warehouse KPIs

Total Stock by Warehouse

Total Orders by Warehouse

Warehouse Utilization
Which warehouse stores the most inventory.

Charts Required for Dashboard:

KPI Cards (Top Section)

- Total Stock
- Total Orders
- Avg Delivery Time
- Stock vs Order Ratio

1. Stock Distribution by Warehouse

Use **Column Chart (not bar)**.

Visual: **Clustered Column Chart**

Axis - Warehouse
Values - StockQuantity (Sum)

This shows **which warehouse stores more inventory**.

2. Supplier Contribution

Use **Donut Chart**.

Legend - Supplier
Values - OrderQuantity (Sum)

This shows **which supplier contributes most orders**.

3. **Delivery Time by Supplier**

Use **Bar Chart**.

Axis - Supplier
Values - Average DeliveryTime

Insight: **Which supplier is fastest**

4. Stock vs Orders Comparison

Use **Clustered Column Chart**

Axis - ProductID
Values -

- StockQuantity
- OrderQuantity

This shows **demand vs available inventory**.

5. **Warehouse vs Orders**

Use **Treemap**

Group - Warehouse
Values - OrderQuantity

This visually shows **which warehouse processes more orders**.

6. Product Demand

Use **Table or Matrix**.

Columns:

- ProductID
- StockQuantity
- OrderQuantity
- DeliveryTime
- Supplier

Sort by **OrderQuantity descending** to see **highest demand products**.